

Coding & Solution

Date	6 July2026
Team ID	
Project Name	Plugging into the Future : An Exploration of Electricity Consumption Patterns
Maximum Marks	5Marks

Coding & Solution

Solution Summary

Field	Details
Repository Link/URL	https://github.com/vinukondadaniel4-sketch/Vinukonda-varshitha-.git
Programming Language(s)	Python3.12,HTML5,CSS3
Framework(s)Used	Flask, Scikit -learn, Pandas, NumPy , Matplotlib , Seaborn
Key Features Implemented	* Interactive Web Dashboard to input household details and instantly view electricity consumption insights. * Machine Learning Model (using Regression/Time-Series) trained to predict future electricity demand & consumption load. * Exploratory Data Analysis (EDA) visualizing peak usage hours, seasonal variations, and high-load periods. * Real-time demand forecasting through a clean Flask web application. * Smart Data Preprocessing (handling missing values, feature scaling) and Feature Engineering (extracting hour, day of the week, month, and weather features) using Pandas and NumPy. * Responsive Frontend designed with modern HTML5 and CSS3 for seamless mobile/desktop usage.
Pending/Incomplete Features	No pending features. All planned functionalities and model deployments have been implemented successfully.

Setup / Run Instructions	1. Clone or download the project repository: git clone https://github.com/vinukondadaniel4-sketch/Vinukonda-varshitha-.git 2. Create a virtual environment: python -m venv venv 3. Activate the virtual environment: .\\venv\\Scripts\\Activate.ps1 (Windows) 4. Install all required packages: pip install -r requirements.txt 5. Navigate to the source folder: Project Development Phase/Source Code 6. Run the local development server: python app.py 7. Open your web browser and visit: http://127.0.0.1:5000
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CodeQualityChecklist

S.No	Criteria	Status(Yes/No)
1	Codeismodularandorganizedintofunctions/classes	Yes
2	Meaningfulvariableandfunctionnamesareused	Yes
3	Codeincludescomments/documentationwherenecessary	Yes
4	Errorhandlingisimplementedforcriticaloperations	Yes
5	Theapplicationrunswithoutcriticalerrors	Yes
6	Codeiscommittedtoaversioncontrolrepository	Yes

AdditionalNotes/Comments:

The Credit Card Approval Prediction system is developed using Python, Flask, and Machine Learning. The application provides an intuitive web interface for users to enter applicant information and instantly receive an approval prediction generated by a trained Random Forest model. The project follows a modular architecture, maintains clean coding standards, and can be easily extended with additional machine learning models or integrated with a database in future enhancements.